

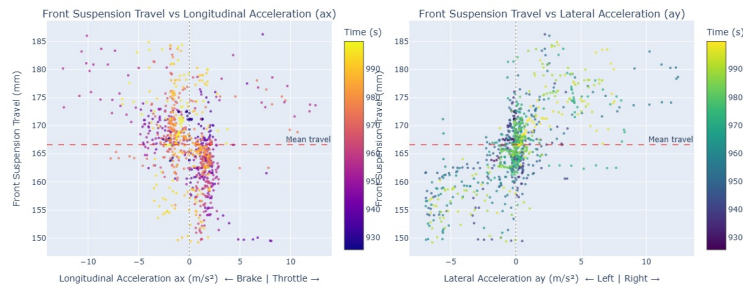
Telemetry Validation

Overview

This report checks our suspension and aero models against real on-track data pulled from the car's IMU and suspension potentiometers.

1. Suspension Travel vs Acceleration

Front suspension travel was plotted against longitudinal acceleration (a_x) and lateral acceleration (a_y) to check the direction and magnitude of load transfer response.

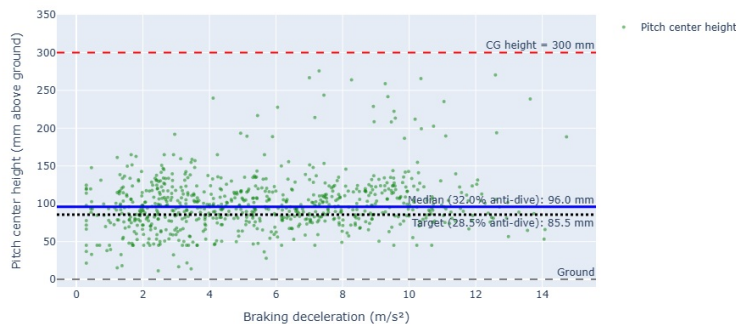


Suspension Travel vs Acceleration

Mean travel is 166.5 mm. Travel vs $|a_y|$ was checked against the expected roll gradient from known wheel rate and roll stiffness to confirm the suspension's real compression matches the kinematic prediction.

2. Pitch Center Height / Anti-Dive

Front travel under braking was regressed against longitudinal deceleration to get a travel-per-decel slope, which was converted to load transfer via the known wheel rate and motion ratio.

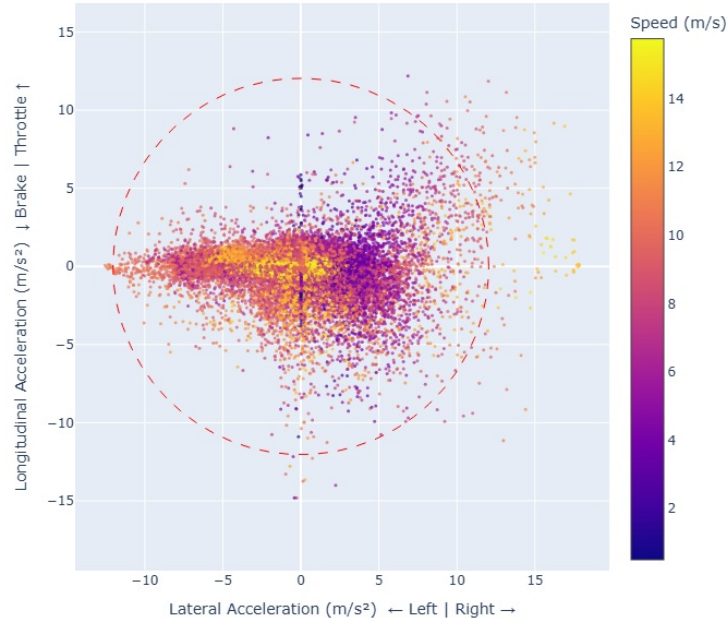


Pitch Center Height

That measured travel was compared against the travel predicted for 0% anti-dive (full load transfer reacted through the spring, with no geometric anti-dive). The shortfall gives pitch center height (96.0 mm median, 32.0% anti-dive) against a target of 85.5 mm (28.5%).

3. GG Diagram

Lateral and longitudinal acceleration were plotted against a reference traction circle to check envelope utilisation across the session.

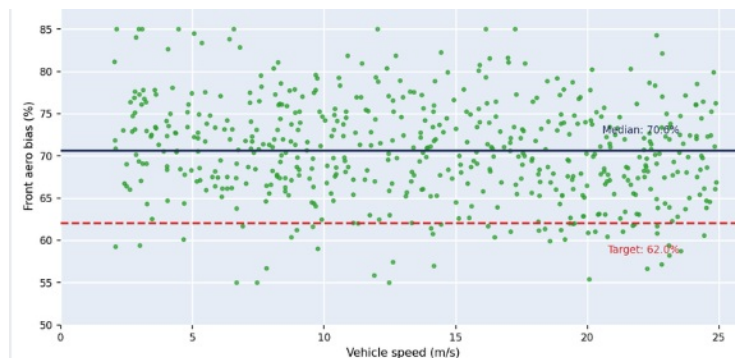


GG Diagram

Peak lateral values approach the traction circle boundary in several corners, and peak braking decel exceeds peak throttle-limited accel, consistent with the car's asymmetric brake/power capability.

4. Front Aero Bias vs Speed

Front bias was computed as front downforce over total downforce, using load estimates derived from ride height and suspension travel at each speed bin.

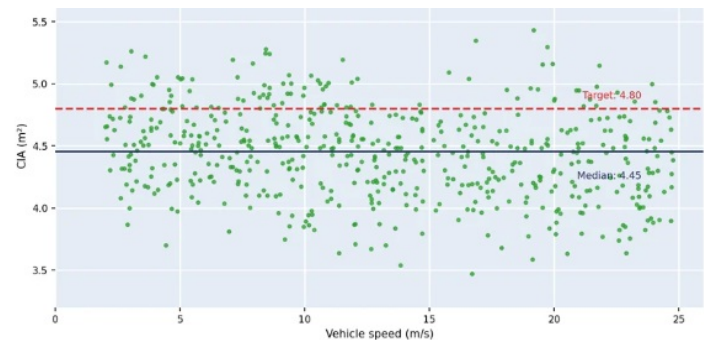


Front Aero Bias

Median bias is 70.0% against a 62.0% target, with no speed dependence, pointing to a fixed offset in the current ride height/aero setup rather than a speed-triggered effect.

5. CIA vs Speed

CIA was back-calculated from measured downforce (via ride height/travel) and dynamic pressure at each logged speed point.



CIA

Median CIA is 4.45 m² against a 4.80 m² target, roughly flat with speed, suggesting a model or setup correction rather than a stall/separation effect.

Summary

Metric	Target	Observed (Median)	Deviation
Anti-dive	28.5%	32.0%	+3.5 pp
Front aero bias	62.0%	70.0%	+8.0 pp
CIA	4.80 m²	4.45 m²	-7.3%